Listening to a Two-Column Page

Reading a two-column paper aloud is an ordering problem before it is a speech problem. The eye follows the left column to its foot and then returns to the top of the right column, while the page itself stores the text in whatever order the producer chose.

Introduction

The reading service must speak the left column first, from its heading to its last line, and only then move to the right column.

A full-width title is not split across the columns. It is spoken once, before either column, and it may become a chapter of its own.

Page numbers and running heads are furniture. They are dropped by the reading-order layout and must never be spoken in the middle of a paragraph.

Method

Each text block is tokenised into words. Every word keeps the rectangles the highlight layer paints while it is being spoken.

Results

The spoken word advances through the sentence, and the amber box follows it. A word split by a hyphen at a line break is spoken once and highlighted on both lines.

A figure has no words to speak. The reader is told that the figure was skipped instead of hearing a description invented for it.

Discussion

Chapters come from the document outline when it has one. The three headings in this paper are also entries in the outline, so the sheet can list them by name.

Figure 1. A figure with no spoken text.